

Towards a Model-based Understanding of Active Cross-situational Word Learning

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Abstract

Word learning is a complex process, influenced not only by the words and referents in the language environment, but also by what learners selectively attend to, which in turn is likely determined by their prior knowledge.

Keywords: active learning; cross-situational word learning; self-directed learning; uncertainty reduction

Introduction

Early word learning is often thought of as largely determined by the learners’ linguistic environment, where they can only learn what is present in the input. Increasingly, research is finding that learners—even at a young age—are involved in selecting and shaping their input, and that such self-directed, active learning can accelerate acquisition. Understanding the strategies that learners use to shape their experience will help us build and improve theories of language.

The goal of our current study is to test how well previously proposed models of cross-situational word learning can explain the observed results of active learning experiments from Zettersten & Saffran (2021). Aside from assessing the ability of the models to account for the overall data, it is of particular interest whether particular model parameters are associated with individuals’ 1) learning performance and 2) deployed active learning strategy. We will also investigate age-related changes in model parameters.

Past work has shown that both children and adults are active word learners, systematically seeking new information word meanings [Hembacher, DeMayo, & Frank (2020); Janakiefski et al. (2022); Zettersten & Saffran (2021); other refs], with important implications for understanding how we are able to quickly build rich vocabularies (Hidaka, Torii, & Kachergis, 2017). Yet what types of active learners are children about word meanings, exactly? While work has begun to characterize a variety of strategies that learners pursue when encountering novel words (George Kachergis, Yu, & Shiffrin, 2013; Zettersten, Cutler, & Lew-Williams, 2023), we lack a formal understanding of the specific sampling strategies that word learners use (Settles, 2012; though see Gelderloos, Mahmoudi Kamelabad, & Alishahi, 2020; Keijser, Gelderloos, & Alishahi, 2019). Recent work has documented broad changes in the sampling strategies that children and adolescents deploy across development across several cognitive domains, e.g. casual learning tasks (Nussenbaum et al., 2019)

and foraging tasks (Meder, Wu, Schulz, & Ruggeri, 2021; Nussenbaum et al., 2023). In the current work, we therefore explore what sampling decision rules best explain both children’s and adults’ information-seeking behavior when learning novel word-referent associations.

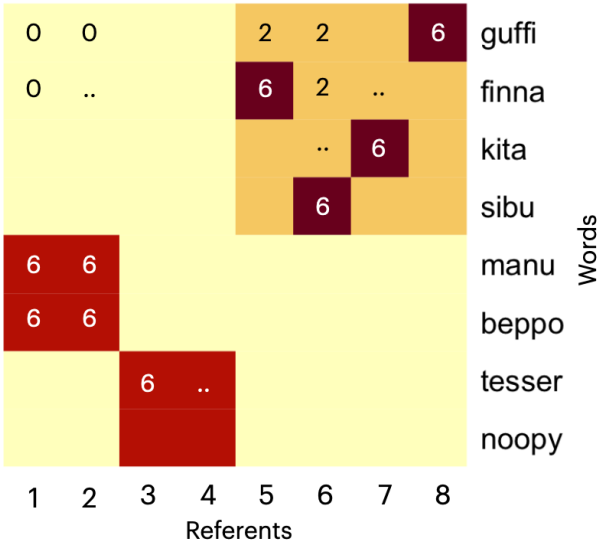


Figure 1: Word-referent co-occurrence matrix for Study (Experiment 1, Zettersten and Saffran, 2020).

Method

Data

All data were taken from Zettersten & Saffran (2021). Data for our Study 1 were pooled from Experiment 1 (31 adults), Experiment 2 (40 children, 4-9 years of age), and adults in the Ambiguous Condition of Experiment S1 (FixMe adults), which had the same design as Experiments 1 and 2.

The training phase of Study 1 contained 24 FixMe-second training trials, each comprised of 2 referents and 2 words. The particular words and referents involved in each pairing were randomized per participant, but the same training trial sequence—and thus co-occurrence statistics—were experienced by all participants. The accumulated word-referent co-occurrences are shown in Figure 1.

Data for Study 2 come from Experiment 3, comprising 58 children of ages 3 to 8.

Models

We focused on analyzing data from the familiarity- and uncertainty-biased associative model G. Kachergis, Yu, & Shiffrin (2016), and variations of that model, described below.

Familiarity- and Uncertainty-biased Associative Model (FUAM) The FUAM assumes that learners accumulate associations between all presented words and referents on each trial, but that their attention is biased more towards particular word-referent associations on the basis of prior experience. In particular, more attention (i.e. associative weight) is directed to associations between words and referents that have appeared together on earlier trials (i.e., a familiarity bias). In addition, more attention is directed to stimuli that have higher uncertainty (i.e. entropy, H) in their associations, whether because they are novel (and thus have no pre-existing associations) or because they have diffuse associations with several different stimuli. The model assumes learners knowledge grows trial-by-trial, and that a fixed amount of associative weight (learning rate χ) is distributed among the presented word-referent associations on a given trial in proportion to the prior familiarity of the association and the uncertainty of the involved word and referent. The relative strength of the familiarity and uncertainty bias is governed by a parameter λ : for values of $\lambda > 1$ the model focuses more attention on uncertain stimuli, while for values of $\lambda < 1$ the model focuses more attention on familiar word-referent associations. Associations are also assumed to decay from trial to trial, at a rate governed by parameter α . The update rule for adjusting the association $M_{w,o}$ between word w and referent o on a given trial is

$$M_{w,o} = \alpha M_{w,o} + \frac{\chi \cdot e^{\lambda \cdot (H(w) + H(o))} \cdot M_{w,o}}{\sum_{w \in W} \sum_{o \in O} e^{\lambda \cdot (H(w) + H(o))} \cdot M_{w,o}} \quad (1)$$

When tested on word w , the model assumes learners choose referent o from the offered alternatives in proportion to their associative strength $M_{w,o}$ (i.e., Luce choice).

The other models we tested are variations on this model.

Familiarity-biased Associative Model (SAM) The familiarity-biased associative model (SAM) implemented just the familiarity bias of the FUAM, without the uncertainty bias term, and thus only implements a bias for already-strong associations, using learning rate (χ) and decay α parameters.

Uncertainty-biased Associative Model (UAM) The uncertainty-biased associative model (UAM) implemented only the uncertainty bias of the FUAM, without the familiarity bias term, and uses the same three parameters as the FUAM.

Novelty-biased Associative Model (NAM) The novelty-biased associative model (NAM), like the UAM, has no famil-

ilarity bias, but instead of the entropy terms ($H(w)$ and $H(o)$) uses novelty (e.g., $1/(frequency(w) + 1)$).

Fitting Procedure

We first fitted each model to each participant's data individually. We then fitted the model simultaneously to all participants in each study. Finally, we again fitted individuals separately, but included a quadratic regularization term in the objective, penalizing parameter values that fell far from the optimal values for the group, approximating a weakly-informative Gaussian prior (-.004..)

Modeling Sampling Choices

Next, we used the best-fit models of participants' learning to investigate participants' search strategies in the sampling phase of the experiment. We considered three sampling decision rules commonly attested in the information-seeking literature to explain active choices: maximizing entropy reduction (information gain), maximizing the probability associated with labels (probability gain), and margin sampling.

Maximizing entropy reduction. In this decision rule, the model's goal is to maximally reduce the entropy of the object-label probability distribution across objects (Nelson, 2005; Settles, 2012). Sampling choices are modeled based on comparing the relative entropy reduction provided by making a given sampling selection of a referent s_o (across the labels w):

$$\max_{s_o, w} (H(w) - H(w|s_o)) \quad (2)$$

Modeling selections were made by determining, on any given trial for a participant, which object maximally reduced entropy (averaging across all possible random distractors that could be chosen in the case of the adult data).

Margin sampling. An alternative strategy documented in describing active sampling strategies is focusing on reducing uncertainty for a more constrained set of items (Markant, Settles, & Gureckis, 2016). This strategy can be formalized as *margin sampling*, in which the model focuses only on the boundary between two alternative options. Margin sampling predicts that learners will focus on instances where there is highest ambiguity between the two most likely labels for a given referent. Formally, if the distribution of labels ($\{l_1, l_2, \dots, l_n\}$) for a given referent o are ordered from most (p_1) to least (p_n) likely, $\{p_1, p_2, \dots, p_n\}$, the margin between labels is given by

$$LM(x) = 1 - (p_1 - p_2) \quad (3)$$

In other words, the model selects the object that has the greatest ambiguity or competition between its two strongest associates.

Maximizing probability gain. In this decision rule, the model's goal is to maximally increase the probability associated between a random label and its object (Nelson, McKen-

zie, Cottrell, & Sejnowski, 2010). Selection choices are modeled based on determining the increase in (correct) association strength provided by a each potential object selection. The final object was selected that led to the largest increase in association strength (averaging across all possible distractors that could be chosen in the case of the adult data).

$$\max_{s_o,w}(M_{w,s_o}) - \max_{w,o}(M_{w,o}) \tag{4}$$

Alternative strategies. We also explored several additional strategies. First, we also considered a *maximum certainty* (or confirmatory) strategy. Learners may gravitate towards confirming choices they already have high certainty for [Watson, 1960; Zettersten et al. (2023)]. Second, we considered a *novelty-based* strategy in which the model based its choices on the relative novelty (or familiarity) of the choice options.

Results

In the individual fits, the median value of $\chi = 0.03$, the median value of $\lambda = 9.25$, and the median value of $\alpha = 0.56$.

In the group fit, the best-fitting parameter values were $\chi = 0.0323$, $\lambda = 3.436$, and $\alpha = 0.895$.

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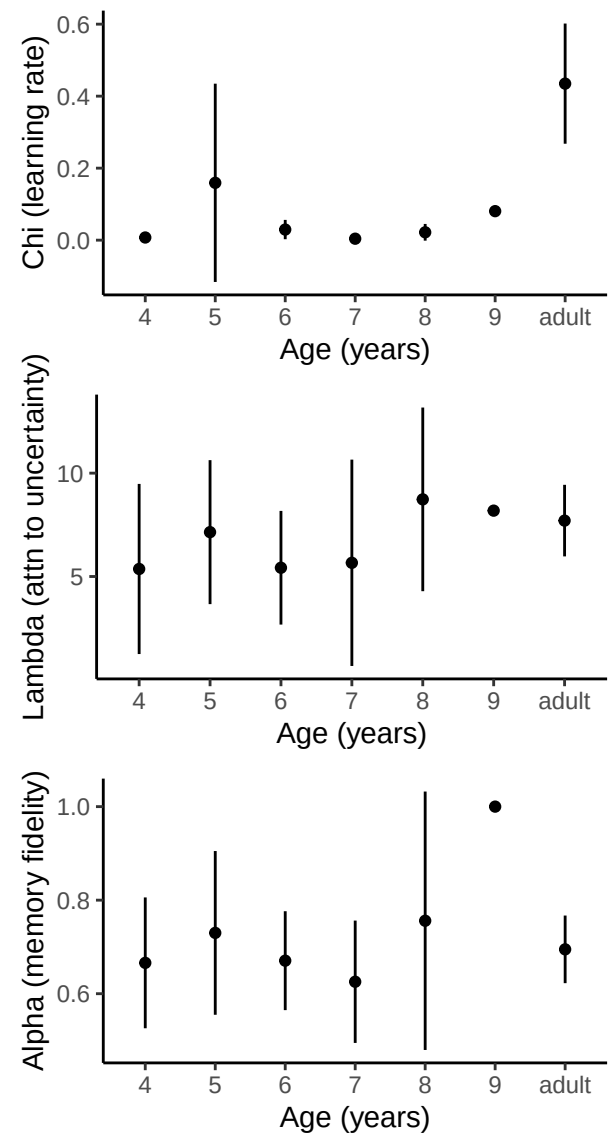


Figure 2: One column image.

Tables

Number tables consecutively; place the table number and title (in 10 point) above the table with one line space above the caption and one line space below it, as in Table 1. You may float tables to the top or bottom of a column, set wide tables across both columns.

You can use the xtable function in the xtable package.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.12	0.09	-1.4	0.16
x	2.11	0.08	26.5	0.00

Table 1: This table prints across one column.

Discussion

Acknowledgements

Place acknowledgments (including funding information) in a section at the end of the paper.

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